Data and Web Science

Poverty Prediction Challenge

Machine Learning 1/2025 – Final Project

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1. Introduction

Measuring poverty accurately is a critical task for policy design and evaluation. However, in many countries, detailed household consumption surveys are conducted infrequently due to high costs, operational challenges, or political instability. As a result, researchers often need to estimate poverty indicators using more recent surveys that lack detailed consumption information. This process is commonly referred to as *survey-to-survey imputation*.

The objective of this project is to predict household-level daily per capita consumption (in 2017 PPP USD) using survey data, and subsequently estimate population-level poverty rates at predefined consumption thresholds. The task is based on a real-world dataset provided by the World Bank through a DrivenData competition and reflects realistic constraints faced in applied poverty monitoring.

2. Dataset Description

The provided dataset consists of household survey data from multiple survey waves. The training set includes three surveys with IDs 100000, 200000, and 300000, while the test set includes three unseen surveys with IDs 400000, 500000, and 600000.

2.1 Files Used

- **train_hh_features.csv**: Household-level survey responses
- **train_hh_gt.csv**: Ground truth household per capita consumption (cons_ppp17)
- **train_rates_gt.csv**: Poverty rates at 19 predefined thresholds for each training survey
- **test_hh_features.csv**: Survey responses for test surveys (without labels)

The datasets were merged using the identifiers *survey_id* and *hhid*. Each household is associated with a population-expanded weight (weight), which reflects both the sampling probability and household size. These weights are used to compute population-level poverty rates.

3. Data Preprocessing

a. *Merging and feature selection*

Household features were merged with the corresponding consumption labels using *survey_id* and *hhid*. Identifier variables were excluded from the model input but retained for submission generation.

b. *Handling missing values*

- Categorical variables (e.g. housing conditions, employment status, food consumption indicators) contained missing values. These were filled with a special category ("__MISSING__") and converted to strings to ensure compatibility with tree-based models.

- Numerical variables were imputed using the median value of each feature.

c. Target transformation

The consumption target variable exhibited a heavily right-skewed distribution. To stabilize training and reduce the influence of extreme values, a logarithmic transformation (log1p) was applied during model training.

d. Feature grouping

The features include demographic characteristics, education and employment indicators, housing and utilities information, geographic variables, and food consumption indicators over the past seven days.

4. Exploratory Data Analysis

Exploratory analysis revealed significant heterogeneity across surveys in both consumption levels and missing data patterns. The distribution of per capita consumption is strongly right-skewed, motivating the use of a log-transformed target. Several variables related to housing quality, employment sector, and food consumption exhibited missing values, reflecting realistic survey limitations.

Overall, the data suggests that poverty is influenced by a combination of household composition, access to basic utilities, employment characteristics, and food consumption patterns.

5. Methodology

5.1 Household Consumption Prediction

The core modeling task is a regression problem, where the goal is to predict daily per capita household consumption. Multiple machine learning models were evaluated, including linear, tree-based, and deep learning approaches.

5.2 Poverty Rate Estimation

Predicted household consumption values were converted into poverty rates using the population weights provided in the dataset. For each survey and poverty threshold t, the poverty rate was computed as:

$$\hat{P}(t) = \frac{\sum_i w_i \cdot 1(\hat{y}_i < t)}{\sum_i w_i}$$

where w_i denotes the household weight and $\hat{y}_i$ the predicted consumption. Poverty was defined strictly as consumption below the given threshold, in accordance with the competition rules.

5.3 Validation Strategy

To ensure realistic evaluation, a leave-one-survey-out cross-validation strategy was adopted. In each fold, two surveys were used for training and the remaining survey for validation. This approach closely mimics the real-world scenario of predicting poverty for an unseen survey.

Performance was assessed using a blended metric that combines household-level consumption error and poverty rate prediction error, with higher weight assigned to poverty rate accuracy.

6. Models Evaluated

The following models were implemented and compared:

- **Ridge Regression**: Linear baseline model
- **Random Forest / Extra Trees**: Non-linear ensemble baseline
- **CatBoost Regressor**: Gradient boosting model optimized for tabular data with categorical features
- **Multilayer Perceptron (MLP)**: Deep learning model trained on tabular features

Hyperparameters were tuned using cross-validation, and early stopping was applied where applicable.

7. Results

Using leave-one-survey-out cross-validation, the CatBoost model achieved a mean blended validation score of 0.0725, combining household-level consumption error and population-level poverty rate error.

Performance varied across surveys, with blended scores of 0.063 for survey 200000, 0.073 for survey 300000, and 0.082 for survey 100000. The poverty-rate prediction error ranged between 3.9% and 5.9%, indicating strong performance on the primary evaluation objective.

The higher error observed for survey 100000 highlights the presence of survey-specific distributional differences, which are expected in real-world poverty monitoring settings.

MEAN blended: 0.07251540723808252

	survey_id	blended	cons_mape	rates_mape	best_iter
0	300000	0.072974	0.278434	0.050146	1359
1	200000	0.062655	0.276769	0.038864	1498
2	100000	0.081917	0.284317	0.059428	1472

8. Submission

New submission

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12.7885

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Submissions

- To help you track your progress during the competition, each submission is scored against publicly available test data to give a "public score".
- **You should select up to 1 submission** to be considered in the final scoring from the table of your submissions that will appear below.
- The primary evaluation metric is a weighted sum of weighted mean absolute percentage error. **Show more**.

Best score	Current rank	Submissions used
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